NeoStats

AI-Powered Credit Risk Intelligence Platform

Intelligence. Innovation. Impact.

LightGBM

ML Model

SHAP

Explainability

Groq LLaMA

Talk-to-Data

Home Credit Default Risk Dataset • Kaggle Competition

PROBLEM & ARCHITECTURE

Business Problem

- Faster, more accurate credit decisions needed
- Explainability required for regulatory compliance
- Analysts need plain-English data exploration
- ~8% default rate with heavy class imbalance

Platform Architecture

Streamlit UI
5 Sections

LightGBM + SHAP
ML & Explainability

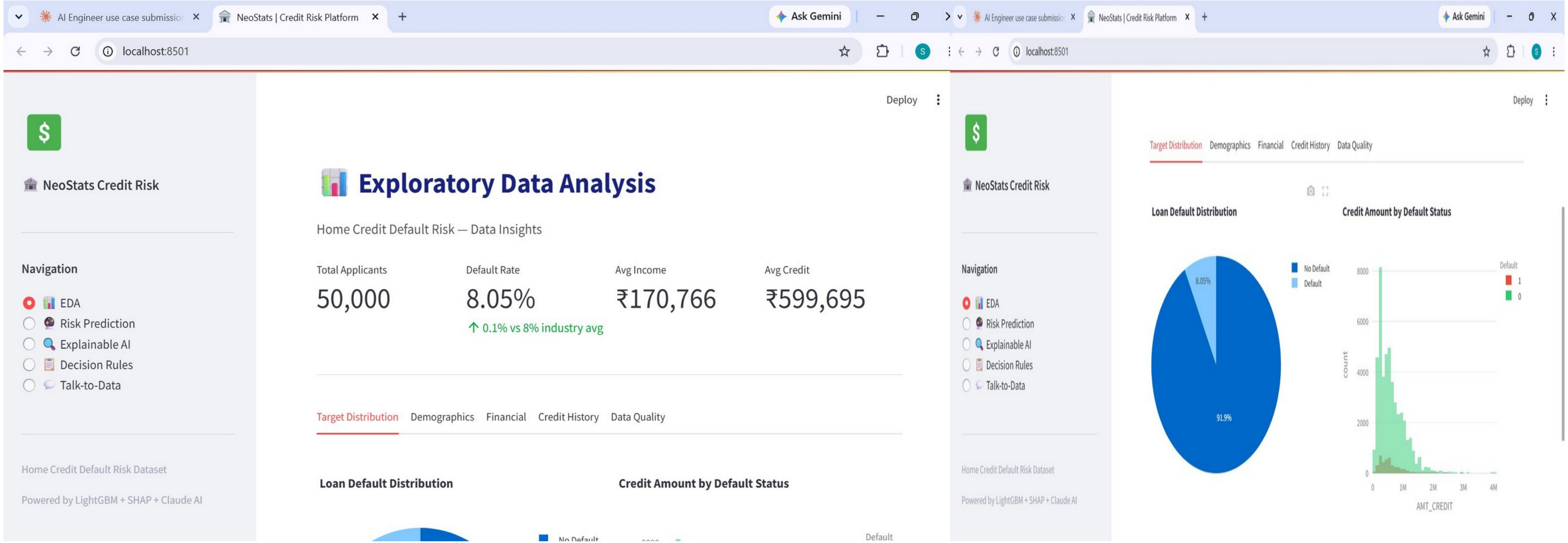
Groq NL→SQL
LLaMA 3.3 70B

SQLite Database
From CSV on init

End-to-end AI engineering: data → model → explainability → chatbot → deployment

EXPLORATORY DATA ANALYSIS

50,000 Applicants	**8.05%** Default Rate	**₹1.71L** Avg Income	**₹5.99L** Avg Credit

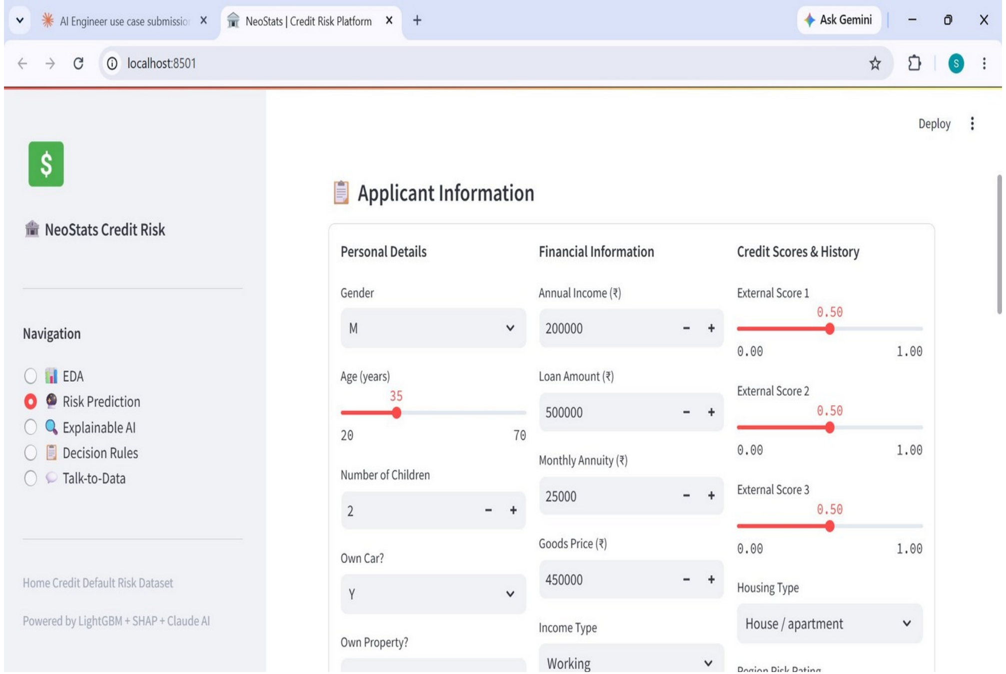


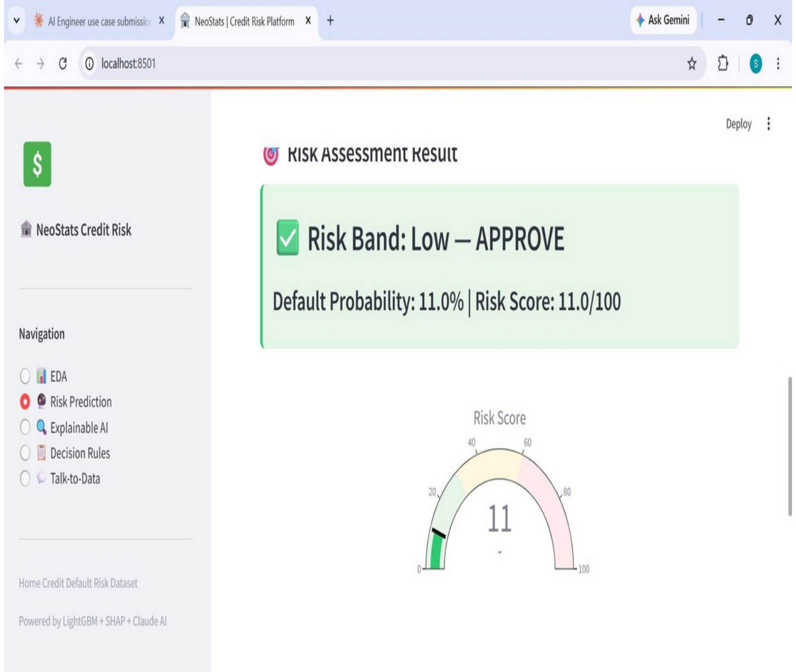
💡 8% imbalance → scale_pos_weight

💡 Younger (20-35) = higher default

💡 EXT_SOURCE 2&3 = top predictors

CREDIT RISK PREDICTION

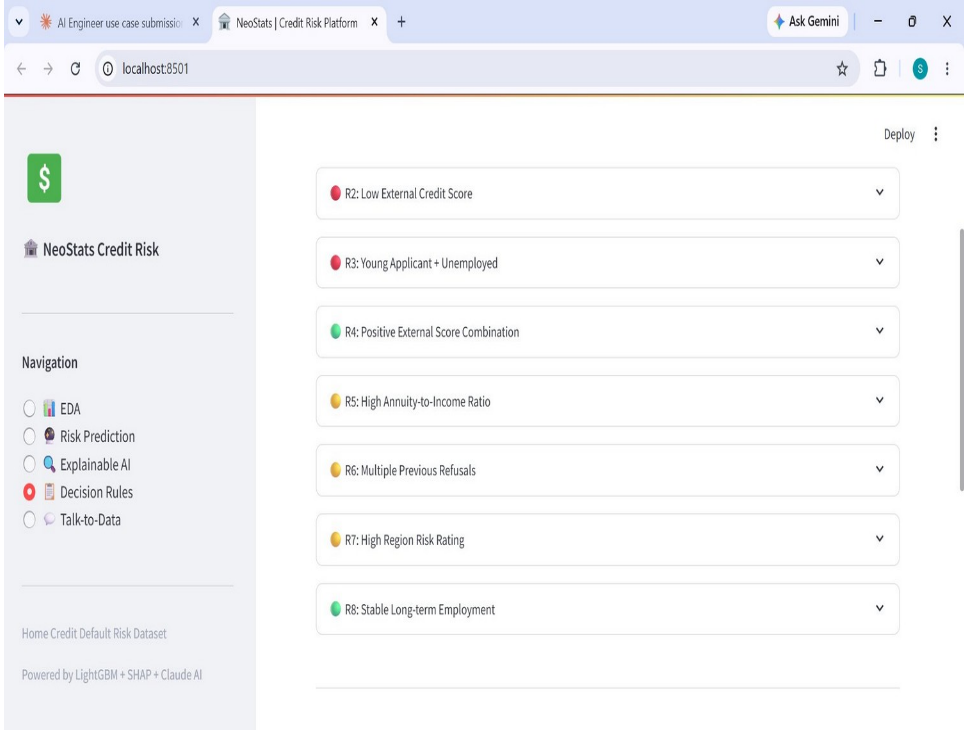




⚡ Model	📊 ROC-AUC	🎯 PR-AUC	⚖️ Imbalance Fix
LightGBM	**0.75–0.77**	**0.35–0.45**	**scale_pos_weight**

Output: Risk Band (Low / Medium / High) + Default Probability + Risk Score (0–100) + SHAP values

BUSINESS DECISION RULES

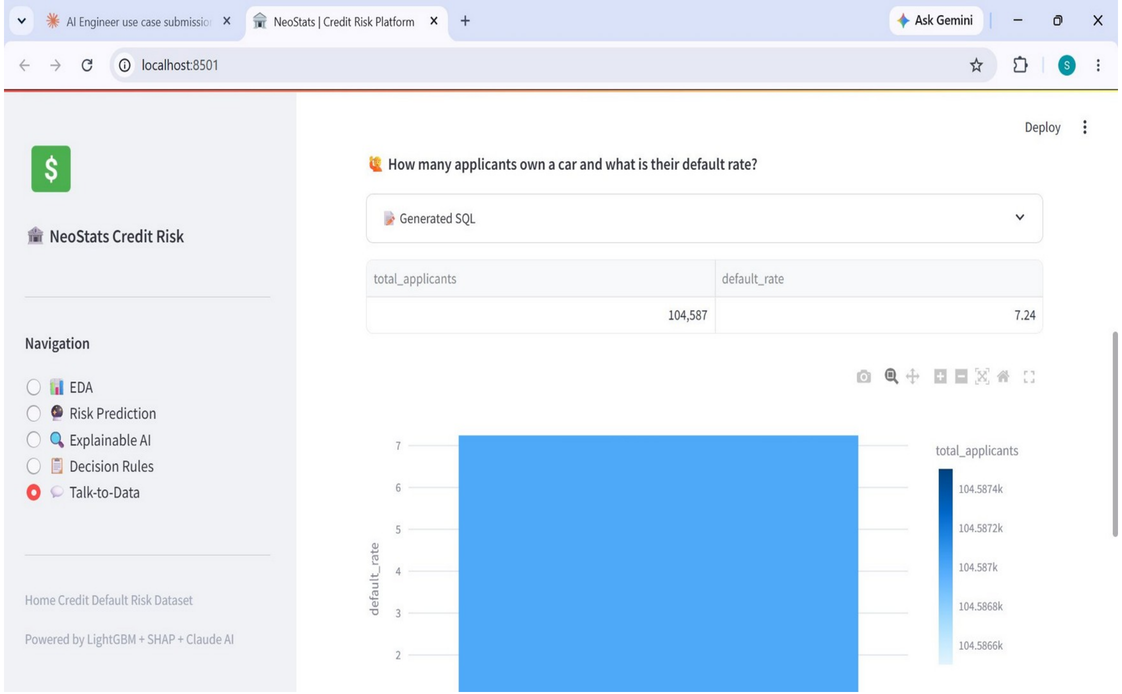


8 ML-Derived Rules

R1	Credit-to-Income > 5×	High
R2	EXT_SOURCE_2 < 0.35	High
R3	Young + Unemployed	High
R4	Triple High Ext Scores	Low
R5	Annuity > 25% Income	Med
R6	2+ Prior Refusals	Med

Rules derived from EDA patterns, SHAP feature importance, and industry credit policy standards

TALK-TO-DATA — NL → SQL CHATBOT



How it works

1. **User types question**
   Plain English query
2. **Groq LLaMA 3.3 70B**
   Generates SQL query
3. **SQL Validation**
   Safety + syntax check
4. **SQLite Execution**
   Runs on real data
5. **Insight Generation**
   Human-readable answer

104,587 car-owning applicants • 7.24% default rate — live query result shown above

Tech Stack & Summary

🖥️ Streamlit

Multi-section UI

🤖 LightGBM

ML + SHAP

💬 Groq LLaMA 3.3

NL→SQL Agent

🗄️ SQLite

Embedded DB

📊 Plotly

Interactive Charts

🐳 Docker

One-command deploy

Full-stack AI solution: EDA → ML → XAI → NL Chatbot → Docker Deployment